

2.156 Challenge Problem 1: Linkage Synthesis

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Code: <https://github.com/Prad-R/2.156-CP1-2026/tree/prads-branch>

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1 Summary

Our final submission scores 3.69 with the course scorer (hypervolumes 5.87, 8.52 and 24.60 for the three kangaroos). The advanced starter notebook, run unchanged, gave us 0.045, so nearly all of that had to come from somewhere else. What ended up working was not a cleverer optimizer but a lot of random guessing followed by ordinary gradient descent, and then a step that lets mechanisms grow one joint at a time. Section 3 goes through these in the order we tried them, Section 4 covers the things we tried that did not work, and Section 5 has the final fronts and mechanisms. The short version of what we could not do: we never managed to trace the tail of Kangaroo 3.

2 The problem as we read it

For each target curve we can submit up to 1000 mechanisms with at most 20 joints. Each one is scored on distance (how far the traced curve is from the target after the best rotation and starting point) and material (total link length), both minimized. A mechanism only counts if it is strictly under both limits in Table 1, and the score per target is the hypervolume with respect to those limits divided by a normalizer. The overall score is the mean of the three.

Problem	Target	Distance limit	Material limit	Normalizer
1	Kangaroo 1 (round body)	0.75	10	2.0
2	Kangaroo 2 (no ears or tail)	1.2	10	1.5
3	Kangaroo 3 (full outline)	1.75	20	10.0

Table 1: Limits and normalizers for the three problems.

Two things about this setup turned out to matter more than we expected. The distance is not normalized for scale, so a curve of the right shape but the wrong size scores badly. On the other hand, scaling every joint position by the same factor scales the curve and the material together, which means each mechanism comes with a free size knob that trades one objective for the other. We use this everywhere. The other thing is that hypervolume is an area: what a design contributes is its gain in distance multiplied by how much material headroom sits above it. A small accuracy gain on a cheap mechanism is worth more than the same gain on an expensive one, which we only really appreciated after stage 4 below.

3 What we did

Table 2 gives the score after each stage. Every score in this report is from `evaluate_submission`.

Stage	What it does	Score after
Starter notebook	Seeded NSGA-II, then gradient descent (Kangaroo 2 only)	0.045
1. Screening	Random mechanisms, pruned and resized, no optimization	2.78
2. Refinement	Adam on joint positions, four material weights per design	3.43
3. Growth	Add one traced joint with two links, refine, select; 20 generations	3.50
4. Wide refinement	About 4000 starts per target, successive halving	3.67
5. Growth again	Same as stage 3, starting from the stage 4 results	3.69

Table 2: Score after each stage.

3.1 Stage 1: guessing a lot

We ran the starter notebook first. From a random start its GA found nothing feasible for a 6-joint mechanism, and seeded with 100 random mechanisms it got to a hypervolume of 0.2 on Kangaroo 2. Our reading of this was that the optimizer was not the bottleneck. The search space of link layouts is huge and most of it is useless, so before optimizing anything we wanted to see what decent mechanisms even look like. We also had access to a cluster, and random sampling is about the easiest thing there is to run in parallel, whereas a GA population has to evolve together.

So the first stage just generates random mechanisms with 10 to 20 joints (each new joint hangs off two earlier ones, as in the course randomizer, with a few random joint layouts per link structure) and solves them in large batches. Two tricks make this far more productive than it sounds. First, we do not pick a traced joint in advance. Every moving joint of every mechanism is treated as a candidate, and for each candidate we throw away all joints that do not drive it, since those only add material. A random 15-joint mechanism therefore gives us a handful of small candidates, and after pruning they had between 4 and 9 joints. Second, the official metric is too slow to run on everything, so candidates are first ranked by a cheap shape comparison done with an FFT (least-squares alignment over all starting points, both directions, rotation and scale). Only the best 40 per target in each batch go to the official metric, and those are scored at nine different sizes so that each one gives several points along its own distance and material trade-off.

Four workers ran this for about four hours each, which came to 20.9 million mechanisms and 22 million official evaluations. With no optimization at all the score was 2.78 (Figure 1). It was already 2.61 after the first worker-hour, so most of those four hours bought very little, and in hindsight we could have stopped much sooner.

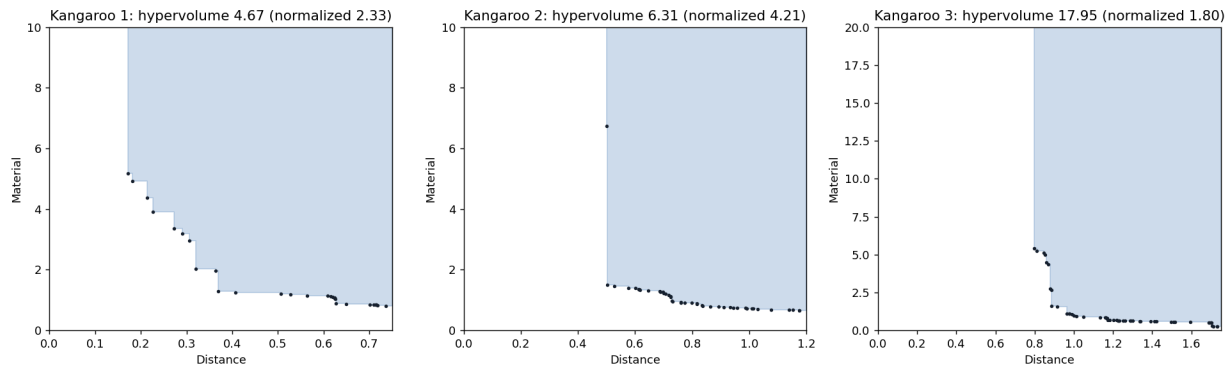


Figure 1: Fronts from random screening alone (score 2.78). The shaded area is the hypervolume. Best distances are 0.17, 0.50 and 0.80.

3.2 Stage 2: gradient descent on the good ones

Next we took the most accurate screened mechanisms for each target and ran Adam on their joint positions for 1500 steps. We picked a fixed number per joint count instead of simply the top N , because otherwise the selection is dominated by one size; this came to 136 mechanisms per target. Each was optimized four times with a different weight on material,

$$\text{distance} + w \cdot \text{material}, \quad w \in \{0, 0.03, 0.1, 0.3\},$$

so that one starting point spreads out into several points on the front. We also kept every intermediate iterate that was inside the limits, not just the last one, which fills in the front for free.

This was the single biggest improvement after screening: 2.78 to 3.43, with best distances down to 0.067, 0.30 and 0.48 (blue curves in Figure 2). Looking at the traced curves at this point, Kangaroo 1 was essentially done. Kangaroo 3 had picked up both ears but its tail and feet were smoothed over, and Kangaroo 2 was rounding off the snout and the feet.

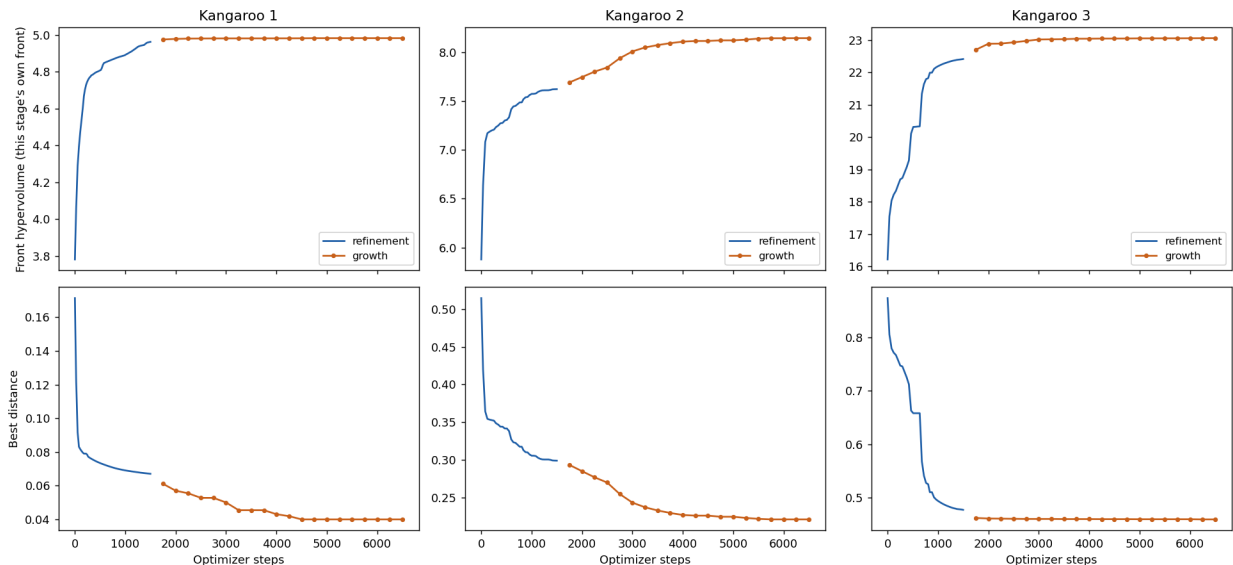


Figure 2: Hypervolume (top) and best distance (bottom) during refinement (blue) and the first growth run (orange, one marker per generation). These are each stage’s own front, not the merged submission. The Kangaroo 1 growth curve is misleading: it ends at a 20-joint design with distance 0.040 that is over the material limit and so does not count. The best valid Kangaroo 1 distance after this stage was 0.054.

3.3 Stage 3: growing mechanisms

Gradient descent can move joints around but it cannot add a link, and a 7-joint mechanism is not going to draw a kangaroo’s ears, tail and feet however well its joints are placed. So we wrote a step that makes mechanisms bigger. Each parent mechanism gets 96 children, and every child is the parent plus one new joint, connected by two links, which becomes the new traced point. The new joint is attached in one of three ways. It can go on the old traced joint and one of that joint’s parents, which puts it on the same rigid body right next to the old tracer, so the child starts out drawing almost the same curve and has one more degree of freedom to improve it. It can go on any two existing joints. Or it can go on one moving joint and a new ground point.

The children are pruned, resized and scored, the best 512 get 250 Adam steps, and the parents for the next generation are the most accurate mechanisms in each joint-count bin (so that small ones are not crowded out by large ones). After 20 generations the score was 3.50 and the best distances were 0.054, 0.22 and 0.46. Kangaroo 2 benefited most, going from 0.30 to 0.22. By the last generation no child was beating its parent on any target.

We made a mistake in this run that is worth mentioning. Parents were selected on distance only, and some of them drifted over the material limit, where nothing they do counts. Kangaroo 1 reached a distance of 0.040 that way with a mechanism we could not submit. We fixed the selection before running growth again.

3.4 Stage 4: a much wider refinement

Two things bothered us about stage 2. It had started from only 136 mechanisms per target, and the best results had come from a few of those that happened to land in a good basin, so it seemed likely that we were leaving good starting points unused. It also had no 4-joint mechanisms in it at all, because they are never among the most accurate, and yet 4-joint mechanisms make up the whole cheap end of every front.

So we reran refinement from about 4000 starting points per target: every seed and front design from screening, everything stages 2 and 3 had produced, and the best designs for the other two kangaroos (the three outlines are variations on one shape, so we expected some transfer). To keep this affordable we used successive halving, with a short run for all 4096, a longer run for the best 1024, and a long run at several material weights for the best 192 plus 64 of the cheap ones.

The score went from 3.50 to 3.67, which was more than growth had given us. What surprised us is that the best distance did not improve on any of the three targets. The entire gain was from finding cheaper mechanisms at accuracies we already had. That is the hypervolume-is-an-area point from Section 2 showing up in practice.

3.5 Stage 5: growth again

Finally we ran the growth step once more, starting from the stage 4 results and with parents kept inside the limits. This brought the score to 3.69. All three design sets in the submitted file come out of this last run.

4 Where the error is, and things that did not work

Once the best distances stopped moving we wanted to know where on the outline the remaining error was, so we aligned our best designs to the target the same way the scorer does and plotted the error at every target point (Figure 3).

Kangaroo 1 has nothing much left to fix. Kangaroo 2 has no single bad region; its most accurate design has several peaks of similar height at the snout tip, the feet and the lower-right corner, while cheaper 6-joint designs miss the concave stretch between snout and feet. Kangaroo 3 is the clear case. Its error is concentrated on the tail, at roughly 55 to 65% of the way around the perimeter, and for the most accurate design the worst fifth of the perimeter accounts for 48% of the distance.

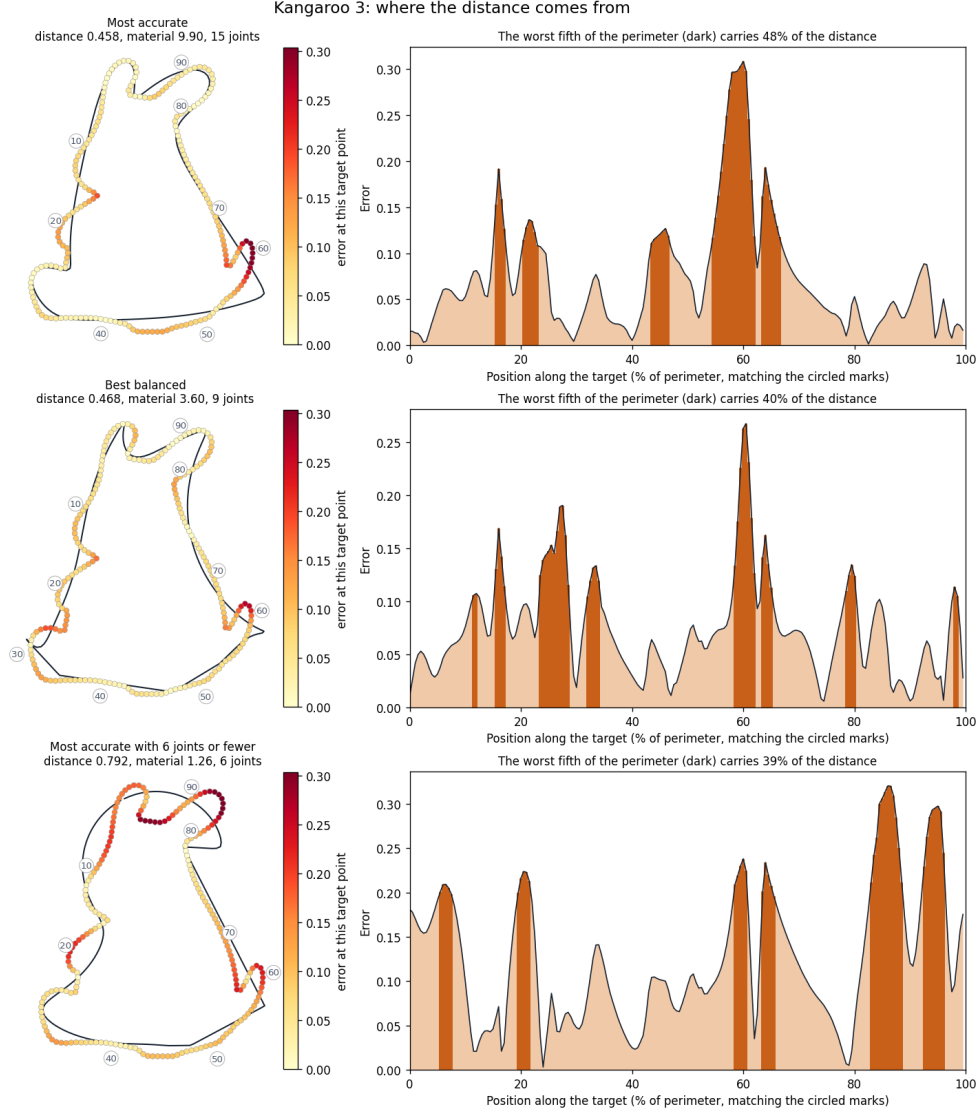


Figure 3: Kangaroo 3, error at each target point (left) and along the perimeter (right) for three designs. This was made partway through, on the evening of October 5, so the numbers differ slightly from the final file. The peak near 60% is the tail.

We think the reason the tail is so hard is in how the metric works. Points are compared at matching positions along the two curves, so a traced curve that pokes halfway into the tail does not get half the credit. It mostly gets its other points pulled out of alignment. From the optimizer's side there is no gentle slope that leads toward drawing a tail.

That led to three experiments, all on existing designs and all trying to lower distance without changing the link layout. None of them worked (Table 3).

Idea	What we did	Hypervolume change (K1 / K2 / K3)
Flipping joints	Mirror a moving joint across the line through its two parents (the other way of assembling the same links), then refine. About 7000 flips.	+0.012 / +0.001 / +0.001
Softer loss	Optimize first on a two-way nearest-point distance, or on a blurred target that is sharpened in steps, then switch to the real distance.	Worse than or equal to a plain control on every target
Careful steps	Only accept a step if it improves one objective without hurting the other.	+0.0007 / +0.0005 / +0.0012

Table 3: Experiments that only move joints.

The softer loss was the one we had hopes for, since it was meant to give partial credit for a partly formed tail. We ran it against a control that used the ordinary loss on the same parents, and the control won: for Kangaroo 3 it reached a hypervolume of 24.36, against 24.27 and 24.19 for the two softer variants. We also learned, a little painfully, that restarting Adam on a design that has already converged tends to knock it out of its minimum and leave it worse than it started, so one has to keep the best point seen and not the final one.

Put together with stage 4, our conclusion is that the designs on our fronts are at real local minima for the link layouts they have. The only thing that was still lowering distance by the end was changing the layout, which is what growth does.

5 Final results

Target	Designs	Hypervolume	Normalized	Best distance	Least material	Joints
Kangaroo 1	1000	5.865	2.933	0.052	0.81	4–14
Kangaroo 2	1000	8.521	5.681	0.219	0.65	4–15
Kangaroo 3	1000	24.604	2.460	0.441	0.27	2–19
Overall score		3.691				

Table 4: The submitted file. As a check we scored it again from a clean copy of the course repository and got the same number.

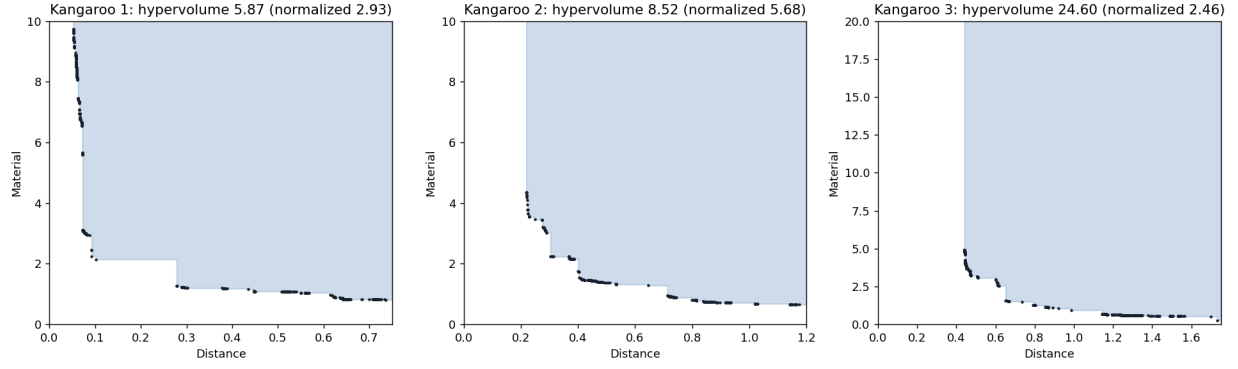


Figure 4: Final fronts, drawn inside their limits.

Figures 5 to 7 show three mechanisms per target: the most accurate one, the lightest one, and what we call the best balanced one, meaning the single design that dominates the largest area on its own.

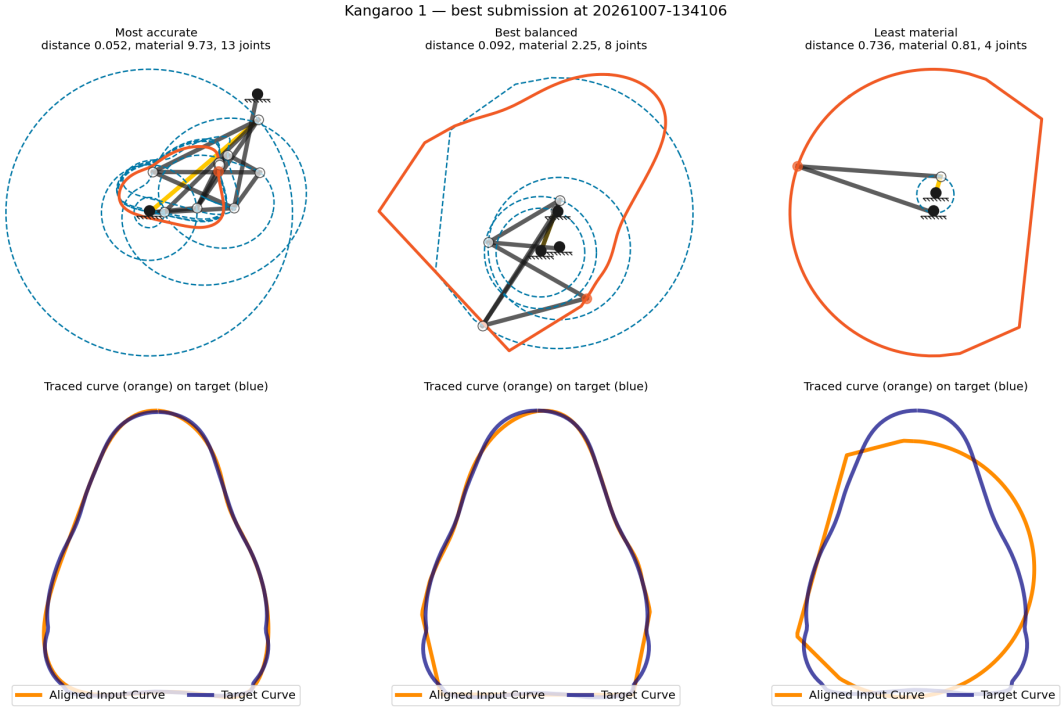


Figure 5: Kangaroo 1. Mechanisms on top, traced curve over the target below.

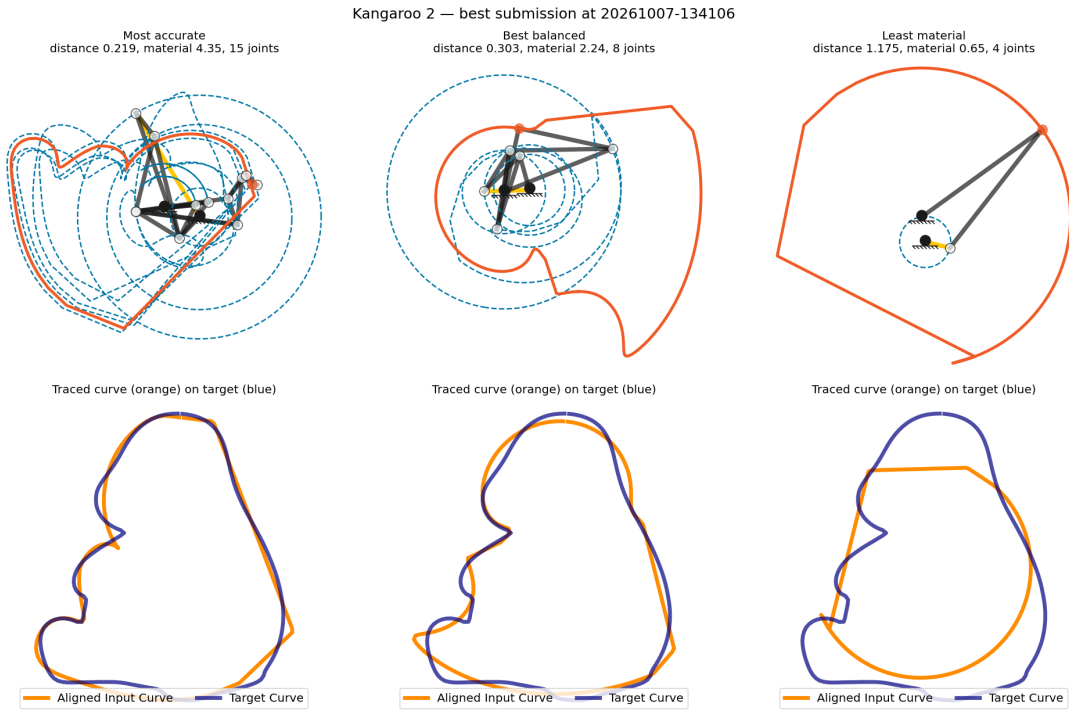


Figure 6: Kangaroo 2.

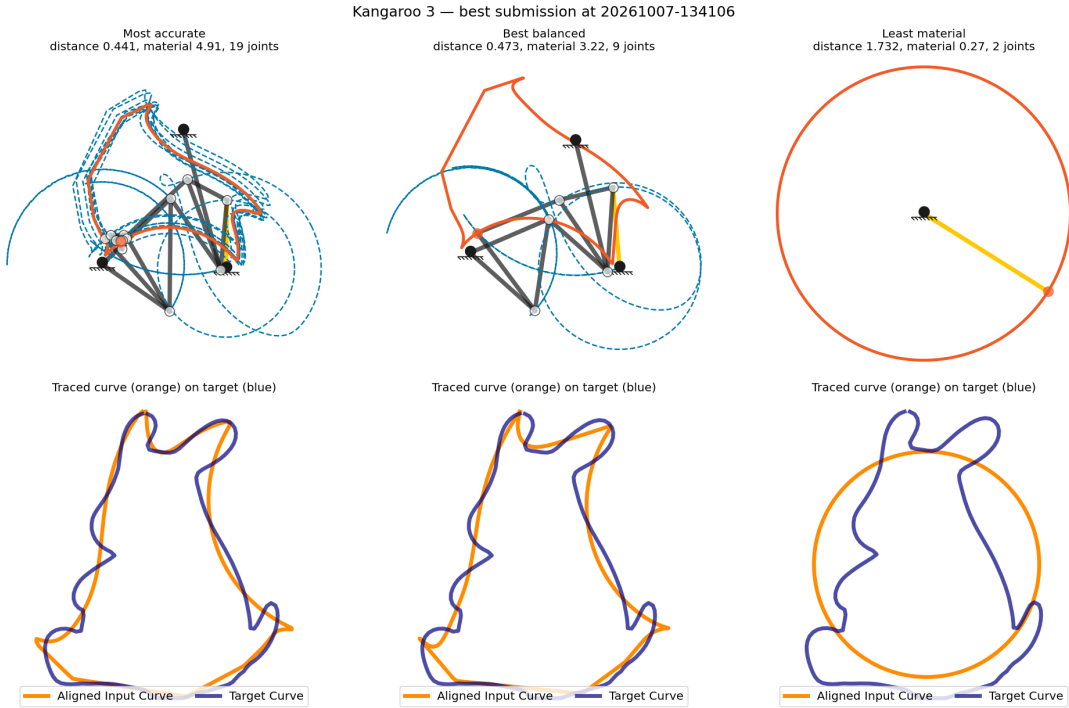


Figure 7: Kangaroo 3. The lightest design is just a crank drawing a circle.

6 Discussion

Looking back at Table 2, random screening alone got us 2.78 out of 3.69 and the first refinement added another 0.65. Everything we did after that, which was most of the effort, added 0.26. If we were short on time we would do stages 1 and 2 and stop.

The fronts for the three targets look quite different. For Kangaroo 1 the most accurate and best balanced designs are close in distance (0.052 and 0.092) but the accurate one uses more than four times the material, so the last bit of accuracy is very expensive. Kangaroo 2 is similar but milder. Kangaroo 3 does not behave like this. Its most accurate and best balanced designs are close in both distance and material, and the front simply stops at about 0.44. It is not the material limit that stops it, since the most accurate design uses 4.91 out of 20. That design does have 19 joints, and we wondered whether the 20-joint limit was what held us back, but the ten joints it has beyond the best balanced design only take the distance from 0.473 to 0.441. We do not think more joints would have fixed the tail. What is missing is the right kind of link layout, and neither random screening nor our growth step found it.

One oddity in the submission is that the lightest Kangaroo 3 design is a bare crank with two joints, whose circle happens to fall just inside the distance limit. The course scorer accepts it. The submission page says mechanisms should have 4 to 20 joints, so it may not be counted there; it accounts for 0.005 of that problem’s hypervolume, so it matters very little either way.

If we had more time we would go after the tail directly. One option is to score growth candidates by their error on the tail stretch of the perimeter instead of the total distance, so that a child that starts to form a tail is kept even if it is worse overall. Another is to search separately for small sub-mechanisms that draw a narrow spike and attach those to the best body we have.

7 Compute and code

Everything ran as CPU jobs on MIT’s Engaging cluster, mostly on the preemptable partition. Jobs there can be killed at any moment, so all our scripts save a checkpoint about once a minute and pick up from it when they restart. Each stage merges its designs with the best submission so far, rescored with the course scorer and keeping the result only if it is better. The code is at <https://github.com/Prad-R/2.156-CP1-2026/tree/prads-branch>; CODE.md there lists what each script does and the commands to rerun the pipeline.